

**UNIVERSITY OF THE FREE STATE
UNIVERSITEIT VAN DIE VRYSTAAT**

BLOEMFONTEIN CAMPUS/KAMPUS

CSIS3744

**DEPARTMENT: COMPUTER SCIENCE AND INFORMATICS
DEPARTEMENT: REKENAARWETENSKAP EN INFORMATIKA**

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**EXAMINATION/EKSAMEN:
Additional End-of-year/Bykomende Jaareind 2017
PAPER/VRAESTEL 1**

ASSESSOR(S)/

ASSESSOR(E): 1. Mnr. W.S.J. Marais

MODERATOR(S)/

MODERATOR(E): 1. Dr. B.P. Haskins

TIME/TYD: 180 min

MARKS/PUNTE: 100

INSTRUCTIONS: Answer all the questions. Number your questions clearly and correctly. The Afrikaans versions of the questions are in italics.

INSTRUKSIES: *Beantwoord al die vrae. Nommer u antwoorde duidelik en korrek. Die Afrikaanse weergawes van die vrae is kursief gedruk.*

Question / Vraag 1 [16 Marks / Punte]

Smith & Wesson Attorneys implemented a 100Base-T Ethernet network with three public wireless access points and an Internet connection. Four administrative computers are connected to the wired network. Smith & Wesson doesn't employ any specialised IT personnel due to limited finances and therefore needs your help with evaluating their network. /

Smith & Wesson Prokureurs het 'n 100Base-T Ethernet netwerk geïmplementeer met drie openbare draadlose toegangspunte en 'n Internet konneksie. Vier administratiewe rekenaars is aan die bedrade netwerk gekoppel. Smith & Wesson het nie enige gespesialiseerde IT-personeel in diens geneem nie as gevolg van beperkte finansies en daarom benodig hulle jou hulp met die evaluering van hulle netwerk.

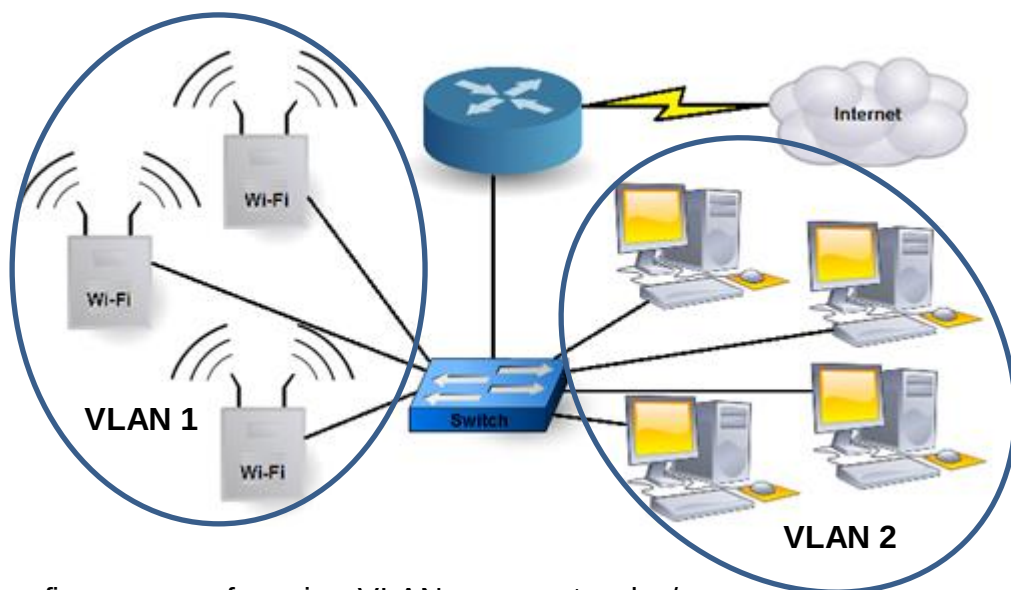
- 1.1 Explain the meaning of the Ethernet standard that has been implemented. /
Verduidelik die betekenis van die Ethernet standaard wat geïmplementeer is. (3)
- 1.2 Which physical media conforms to the Ethernet standard for this network? /
Watter fisiese media voldoen aan die Ethernet standaard vir hierdie netwerk? (1)
 - 1.2.1 Discuss three advantages with regard to this type of physical media. /
Bespreek drie voordele ten opsigte van hierdie tipe fisiese media. (3)

- 1.3 Give two reasons why Smith & Wesson implemented a peer-to-peer network rather than a client/server network. /
Gee twee redes waarom Smith & Wesson eerder 'n eweknie-netwerk geïmplementeer het as 'n kliënt/bediener-netwerk. (2)
- 1.4 Explain to Smith & Wesson why it would be better to set up a client/server network by comparing the disadvantages of peer-to-peer networks with the advantages of client/server networks. /
Verduidelik aan Smith & Wesson hoekom dit beter sal wees om 'n kliënt/bediener-netwerk op te stel deur die nadele van eweknie-netwerke met die voordele van kliënt/bediener-netwerke te vergelyk. (7)
 [16]

Question / Vraag 2 [27 Marks / Punte]

Smith & Wesson Attorneys' current network address is 192.168.0.0. Smith & Wesson wants to separate its administrative computers from the wireless network by dividing the network into two VLANs as illustrated in the following network diagram: /

Smith & Wesson Prokureurs se huidige netwerkadres is 192.168.0.0. Smith & Wesson wil die administratiewe rekenaars skei van die draadlose netwerk deur die netwerk in twee VLAN's op te deel soos geïllustreer in die volgende netwerkdiagram:



- 2.1 Give five reasons for using VLANs on a network. /
Gee vyf redes vir die gebruik van VLAN's op 'n netwerk. (5)
- 2.2 What type of connectivity device would the administrator have to reconfigure in order to set up the VLANs? /
Watter tipe verbindingstoestel sal die administrateur moet verstel ten einde die VLAN's op te stel? (1)
- 2.3 Explain how trunking protocols operate.
Verduidelik hoe stamprotokolle funksioneer. (2)
- 2.3.1 Why is Cisco's VLAN trunking protocol the most popular trunking protocol? /
Hoekom is Cisco se VLAN-stamprotokol die gewildste stamprotokol? (2)

- 2.4 Specify how many collision domains and how many broadcast domains Smith & Wesson's network will contain when the VLANs are implemented. /
Spesifiseer hoeveel botsing-domeine en hoeveel uitsaai-domeine Smith & Wesson se netwerk sal bevat nadat die VLAN's geïmplementeer word. (2)
- 2.4.1 Motivate your answer by stipulating which portion of the network falls within the broadcast domain(s) and which portion falls within the collision domain(s). /
Motiveer jou antwoord deur te stipuleer watter deel van die netwerk binne die uitsaai-domein(e) val en watter deel in die botsing-domein(e) val. (2)
- 2.5 Discuss the potential problem that is created when a network is divided into VLANs and how this problem can be solved. /
Bespreek die potensiele probleem wat veroorsaak word wanneer 'n netwerk in VLAN's verdeel word en hoe hierdie probleem opgelos kan word. (3)
- 2.6 Smith & Wesson Attorneys also need to assign each VLAN its own subnet of IP addresses by creating two subnets. /
Smith & Wesson Prokureurs moet ook aan elke VLAN sy eie subnetwerk van IP-adresse toeken deur die skep van twee subnetwerke.
- 2.6.1 Calculate the new subnet mask for Smith & Wesson Attorneys. /
Bereken die nuwe subnetmasker vir Smith & Wesson Prokureurs. (5)
- 2.6.2 How many mobile devices will VLAN 1 be able to host? /
Hoeveel mobiele toestelle sal VLAN 1 in staat wees om te huisves? (1)
- 2.7 Complete the table below by supplying each subnet's network address, broadcast address and usable host address range: /
Voltooi die onderstaande tabel deur elke subnetwerk se netwerkadres, uitsaai-adres en bruikbare gasheer-adresreeks te verskaf:

	Subnet	Network Address <i>Netwerkadres</i>	Broadcast Address <i>Uitsaai-adres</i>	Usable Host Address Range <i>Bruikbare Gasheer-adresreeks</i>
2.7.1	1			
2.7.2	2			

(2)

(2)

[27]

Question / Vraag 3 [20 Marks / Punte]

Smith & Wesson Attorneys' wireless access points are configured as follows: /
Smith & Wesson Prokureurs se draadlose toegangspunte is as volg opgestel:

- Access point 1's SSID: SandW1WiFi / *Toegangspunt 1 se SSID: SandW1WiFi*
- Access point 2's SSID: SandW2WiFi / *Toegangspunt 2 se SSID: SandW2WiFi*
- Access point 3's SSID: SandW3WiFi / *Toegangspunt 3 se SSID: SandW3WiFi*
- Encryption standard: WEP / *Enkripsiestandaard: WEP*
- DHCP: Enabled / *DHCP: Aangeskakel*
- Wi-Fi Protected Setup: Disabled / *WiFi Beskernde Opstelling: Afgeskakel*
- Supported 802.11 standards: n/ac / *Ondersteunde 802.11 standaarde: n/ac*

- 3.1 Identify and explain all the relevant security risks that are contained in the access points' configuration settings and give a solution for the vulnerabilities. / *Identifiseer en verduidelik al die tersaaklike veiligheidsrisiko's vervat in die toegangspunte se konfigurasie-instellings en gee 'n oplossing vir die kwesbaarhede.* (2)
- 3.2 Name and discuss the protocol that is implemented by wireless access points to counter the "hidden node problem". / *Noem en bespreek die protokol wat deur draadlose toegangspunte geïmplementeer word om die "verborge node problem" teen te werk.* (4)
- 3.3 Compare all five Wi-Fi standards in a table with regard to frequency band and maximum theoretical throughput. / *Vergelyk al vyf Wi-Fi standaarde in 'n tabel ten opsigte van frekwensieband en maksimum teoretiese deursetvermoë.* (7)
- 3.4 The throughput of the wireless LAN drops to 4 Mbps when there is more than 30 users. Describe the probable cause for the low throughput and explain how this pitfall can be avoided. / *Die deursetvermoë van die draadlose LAN daal na 4 Mbps wanneer daar meer as 30 gebruikers is. Beskryf die waarskynlike oorsaak vir die lae deursetvermoë en verduidelik hoe hierdie slagkat vermy kan word.* (3)
- 3.5 Discuss the technologies, which allow 802.11n/ac to achieve such high throughput rates. / *Bespreek die tegnologieë, wat 802.11n/ac toelaat om so hoë deursetkoerse bereik.* (4)
- [20]

Question / Vraag 4 [22 Marks / Punte]

Smith & Wesson Attorneys has an ADSL Internet connection. / *Smith & Wesson Prokureurs het 'n ADSL Internet konneksie.*

- 4.1 Identify the router that connects Smith & Wesson's LAN to the Internet; is it an interior, exterior or border router? / *Identifiseer die roetebepaler wat Smith & Wesson se LAN aan die Internet verbind; is dit 'n interieure-, eksterieure- of grensroetebepaler?* (1)
- 4.1.1 On what level of the OSI model does a router operate? / *Op watter vlakke van die OSI-model funksioneer 'n roetebepaler?* (1)
- 4.1.2 Discuss the OSPF link-state routing protocol. / *Bespreek die OSPF skakel-toestand roetebepalingsprotokol.* (4)
- 4.2 Explain in detail how an ADSL connection is established between a home computer and the ISP. / *Verduidelik deeglik hoe 'n ADSL-verbinding geskep word tussen 'n huisrekenaar en die ISP.* (5)
- 4.3 What protocol is used for connecting home computers to an ISP via DSL? / *Watter protokol word gebruik vir die aansluiting van huisrekenaars aan 'n ISP via DSL?* (1)

- 4.4 What server is necessary for users to connect to a LAN from a remote location using the Point-to-Point Protocol (PPP)? /
Watter bediener is nodig vir gebruikers om by 'n LAN aan te sluit van 'n afgeleë plek deur die Punt-na-Punt-Protokol (PPP) te gebruik? (1)
- 4.4.1 Why is Point-to-Point Protocol (PPP) a better protocol than Serial Line Internet Protocol (SLIP)? /
Waarom is Punt-tot-Punt Protokol (PPP) 'n beter protokol as Serie-Lyn Internet Protokol (SLIP)? (5)
- 4.5 Discuss the difference between a site-to-site VPN and a client-to-site VPN. /
Bespreek die verskil tussen 'n "site-to-site" VPN en 'n "client-to-site" VPN. (4)
 [22]

Question / Vraag 5 [15 Marks / Punte]

The DNS server, dns05.server.net, has the following IP address: 146.182.15.25. Its IP address was assigned by the DHCP server. /

Die DNS-bediener, dns05.server.net, het die volgende IP-adres: 146.182.15.25. Sy IP-adres is deur die DHCP-bediener toegeken.

- 5.1 Discuss the steps involved in the DHCP leasing process. /
Bespreek die stappe wat betrokke is met die DHCP verhuringsproses. (4)
- 5.2 Write down the socket address that is used to request DNS information from the server. /
Skryf die sokadres neer wat gebruik word om DNS inligting van die bediener af aan te vra. (2)
- 5.3 What console command can be used to determine whether the DNS server is responding? Write down the full command. /
Watter konsole-instruksie kan gebruik word om te bepaal of die DNS-bediener reageer? Skryf die volledige instruksie neer. (2)
- 5.3.1 Explain how this command uses ICMP and ARP to perform its task. /
Verduidelik hoe hierdie instruksie ICMP en ARP gebruik om sy taak uit te voer. (3)
- 5.4 What domain does the DNS server belong to? /
Aan watter domein behoort die DNS-bediener? (1)
- 5.5 What is the host name of the DNS server? /
Wat is die gasheernaam van die DNS-bediener? (1)
- 5.6 Briefly explain the difference between unicast, and multicast IPv6 addresses. /
Verduidelik kortliks die verskil tussen "unicast", en "multicast" IPv6-adresse. (2)
 [15]

End of Paper / Einde van Vraestel